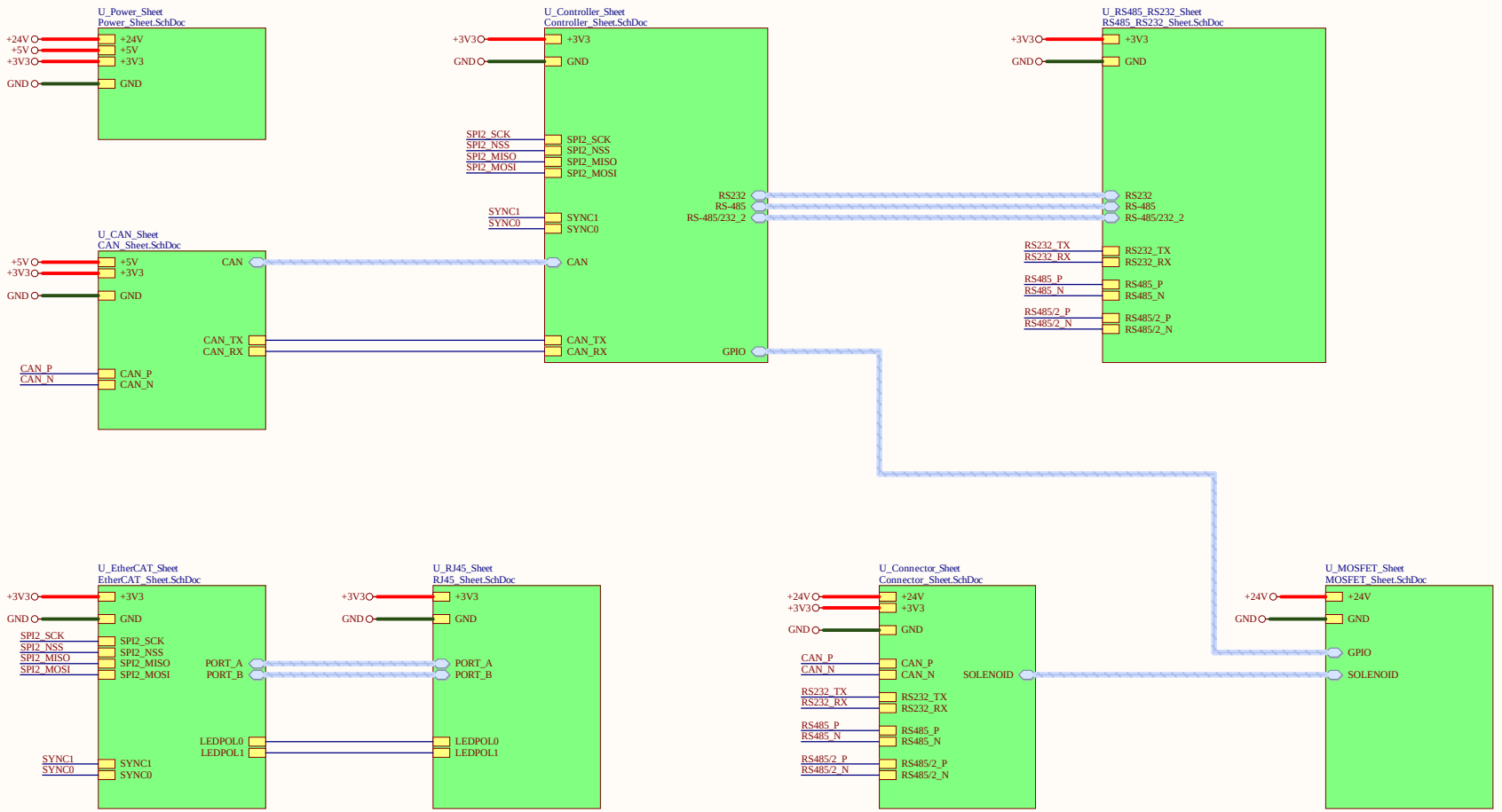
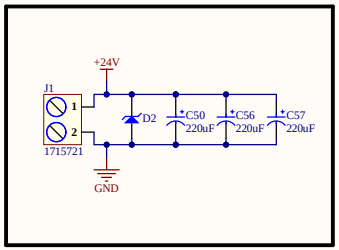


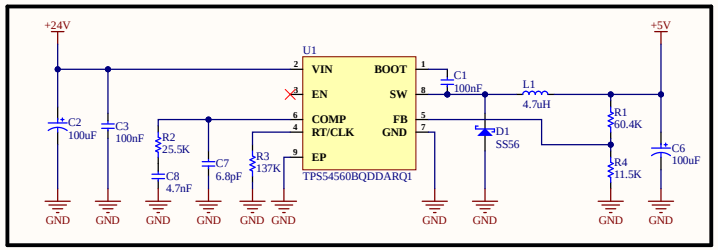
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Title: Top_Sheet.SchDoc	
File: Solenoid-Switching-Board-v1.0.PrjPcb	
Date: 8/7/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 9

24V INPUT

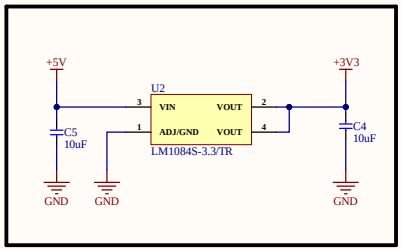


5V 5A BUCK



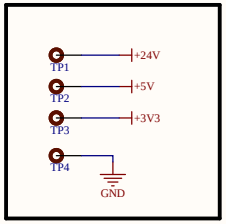
5A

3.3V 3A LDO

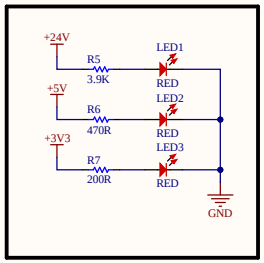


3A

TEST POINTS



LEDs

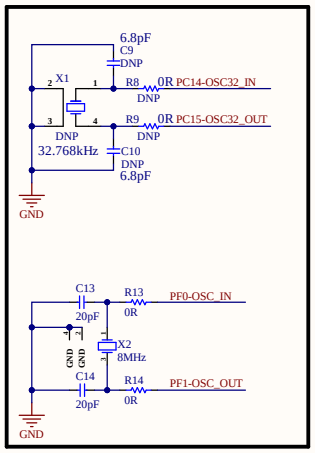




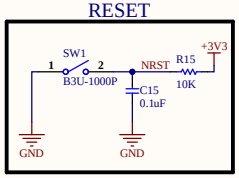
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Drawn By: Huzaifa Ali	Page	Total: 9

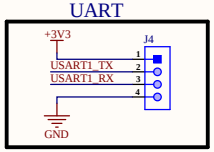
CRYSTAL



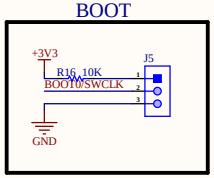
RESET



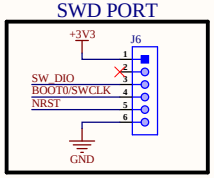
UART

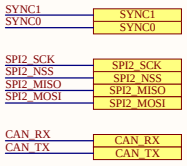


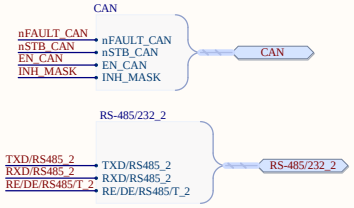
BOOT

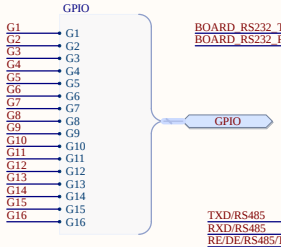


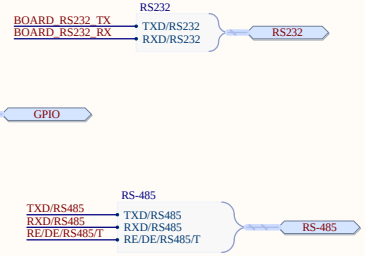
SWD PORT





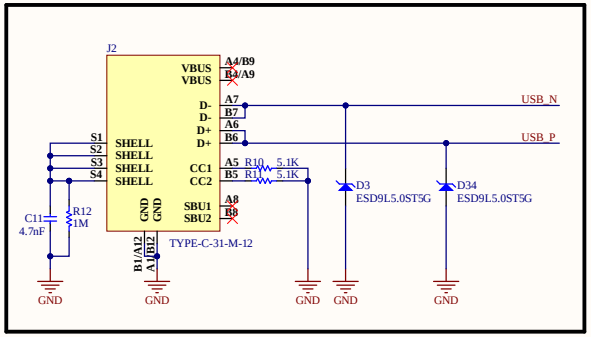




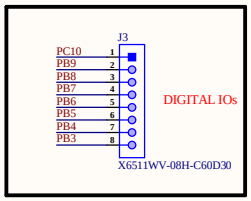




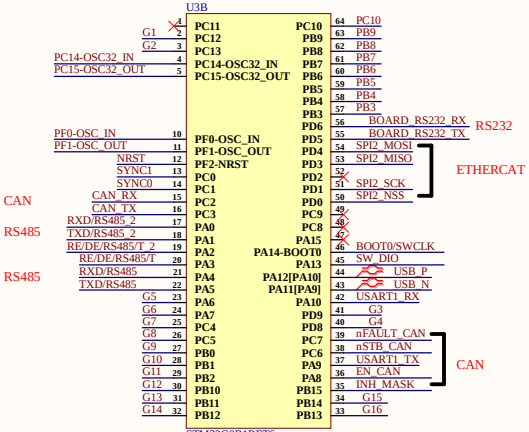
USB

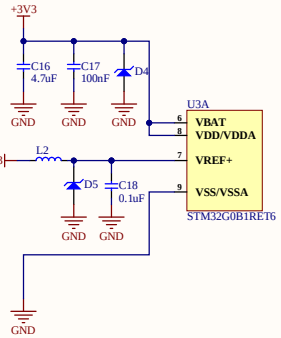


DIO ADC



DIGITAL IOs REQUESTED BY CLIENT

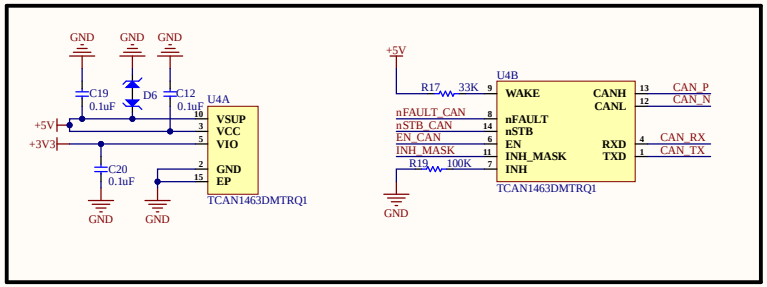




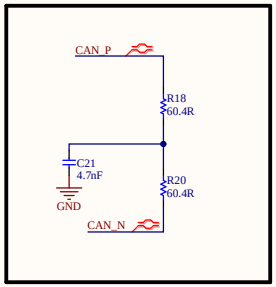
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Date: 8/19/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 9

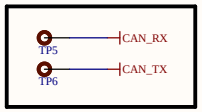
CAN BUS



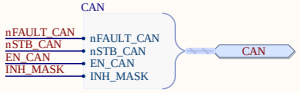
CAN TERMINATION

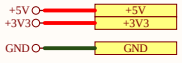


TESTPOINTS





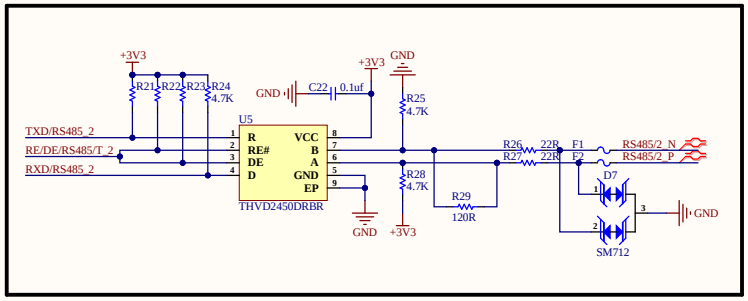




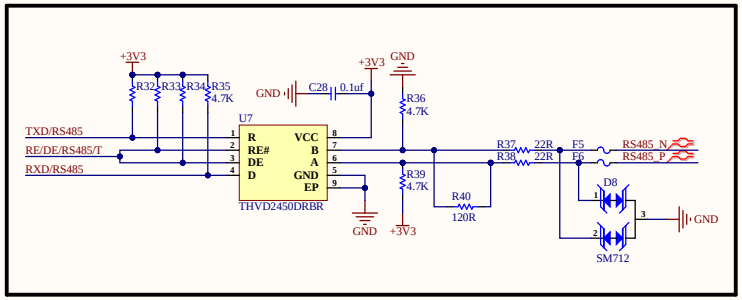
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File: Solenoid-Switching-Board-v1.0.PrjPcb	
Date: 8/13/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 9

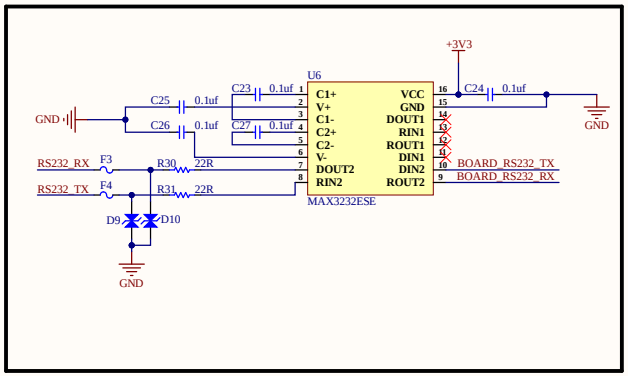
RS485_2 TO UART



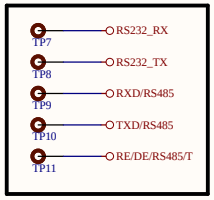
RS485 TO UART

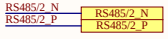


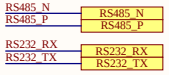
RS232 TO UART



TESTPOINTS









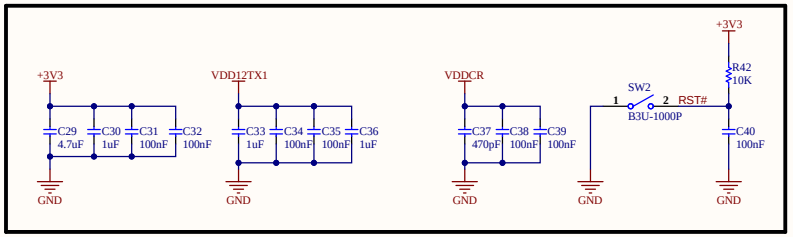


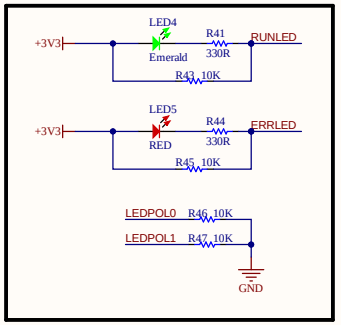




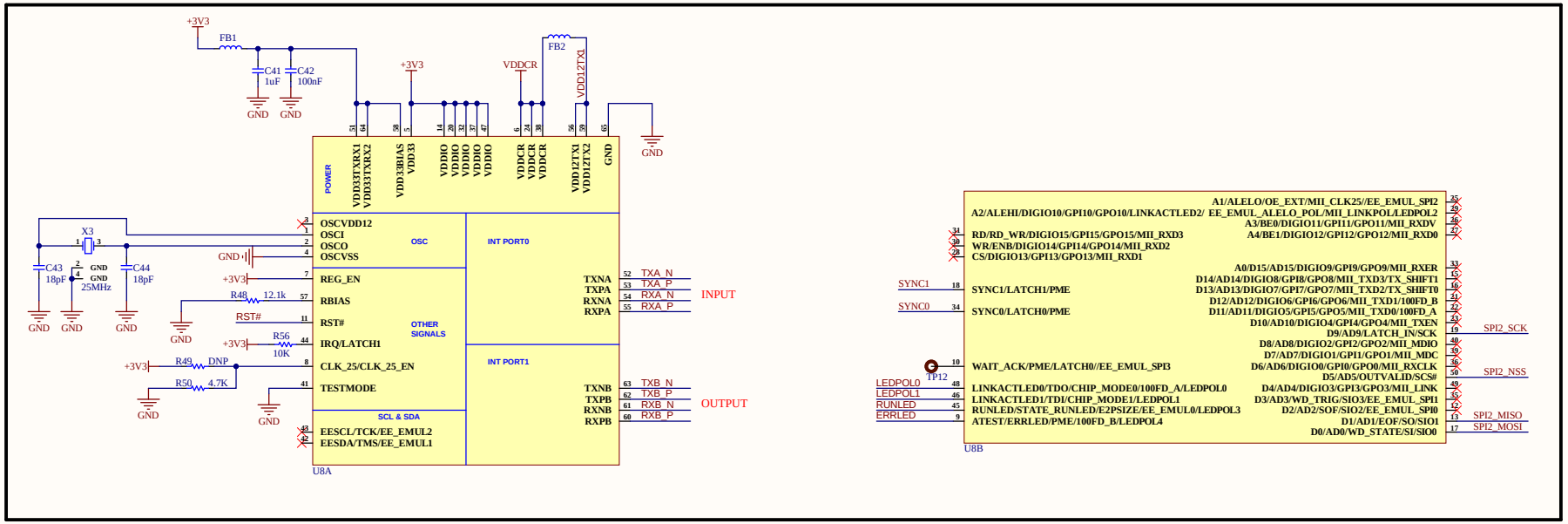
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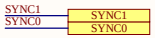
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Date: 8/16/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 9

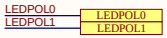




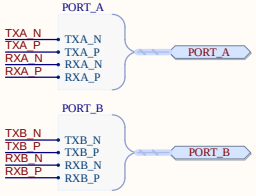
ETHERCAT









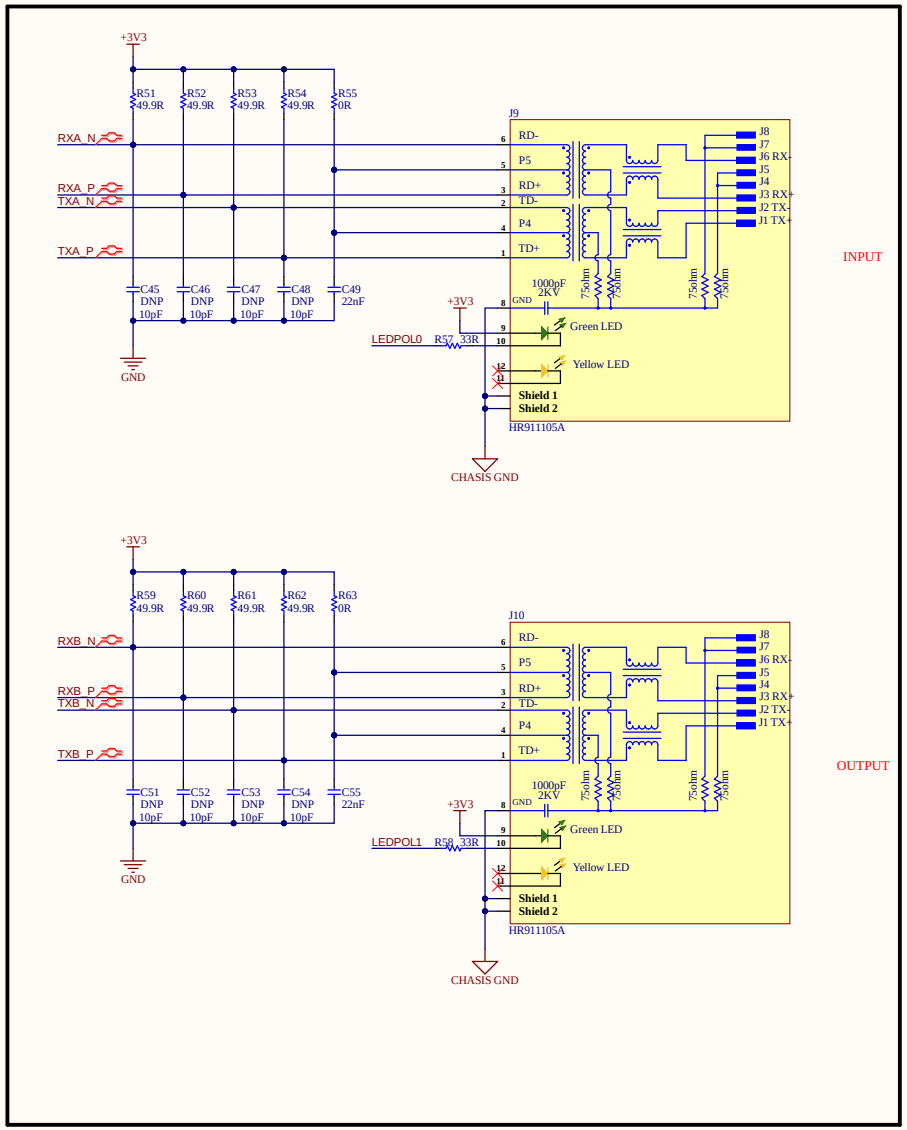




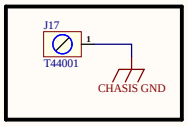
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File: Solenoid-Switching-Board-v1.0.PrjPcb	
Date: 8/15/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

RJ45



EARTH TERMINAL



LED1 (Green) = LINK/ACT
LED2 (Yellow) = SPEED

PORT_A
TXA_N
TXA_P
RXA_N
RXA_P
PORT_A

PORT_B
TXB_N
TXB_P
RXB_N
RXB_P
PORT_B

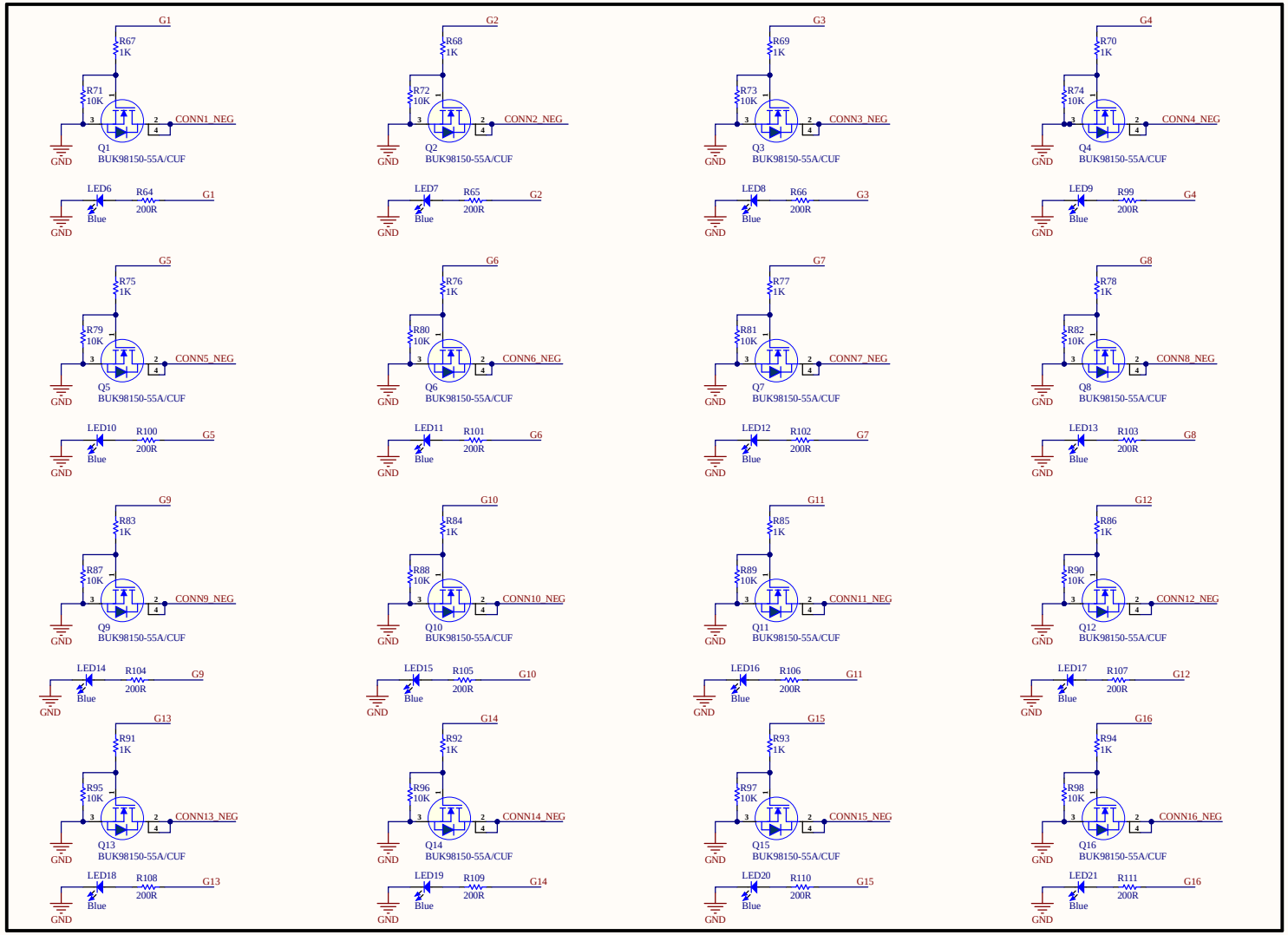
LEDPOL0
LEDPOL1

+3V3
GND

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Title: RJ45_Sheet.SchDoc	
File: Solenoid-Switching-Board-v1.0.PrjPcb	
Date: 8/7/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 9

SOLENOID DRIVER



MAKE SECTIONS OF 4x4

SOLENOID

CONN1_NEG → CONN1_NEG
CONN2_NEG → CONN2_NEG
CONN3_NEG → CONN3_NEG
CONN4_NEG → CONN4_NEG
CONN5_NEG → CONN5_NEG
CONN6_NEG → CONN6_NEG
CONN7_NEG → CONN7_NEG
CONN8_NEG → CONN8_NEG
CONN9_NEG → CONN9_NEG
CONN10_NEG → CONN10_NEG
CONN11_NEG → CONN11_NEG
CONN12_NEG → CONN12_NEG
CONN13_NEG → CONN13_NEG
CONN14_NEG → CONN14_NEG
CONN15_NEG → CONN15_NEG
CONN16_NEG → CONN16_NEG

SOLENOID

GPIO

G1 → G1
G2 → G2
G3 → G3
G4 → G4
G5 → G5
G6 → G6
G7 → G7
G8 → G8
G9 → G9
G10 → G10
G11 → G11
G12 → G12
G13 → G13
G14 → G14
G15 → G15
G16 → G16

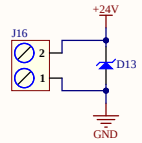
GPIO

+24V → +24V
GND → GND

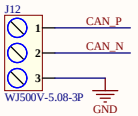
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Date: 8/9/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 9

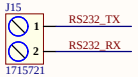
+24V OUTPUT



CAN



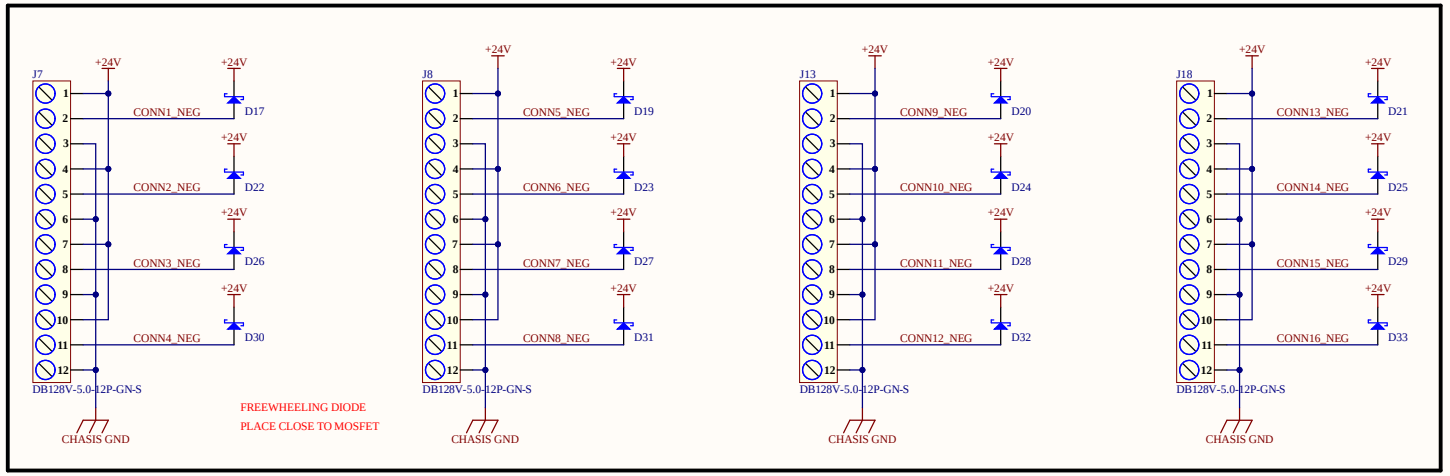
RS-232

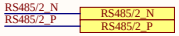


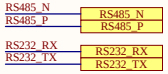
RS-485

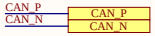


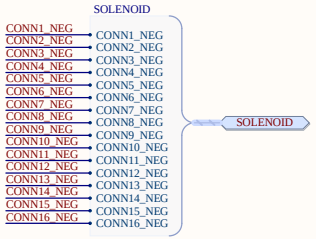
OUTPUT CONNECTORS













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File: Solenoid-Switching-Board-v1.0.PrjPcb	
Date: 8/9/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 9